AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Smart Bicycle Lock with Mechanical Backup

Team ID: *GEN26-A0H-I069*

Team Name: *ASGUARDIANS*

Domain: *Energy, Mobility & Smart Infrastructure*

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Problem Statement

Bicycle owners need a secure and convenient way to protect their bicycles when parked. Conventional locks depend on physical keys or combinations, while electronic locks may become inaccessible when power or electronic components fail. The challenge is to combine smart access with an independent mechanical backup that works without electrical power.

Problem Understanding

- **Target users:** Students, commuters and bicycle owners using campus or public parking.
- **Pain points:** Theft, tampering, lost keys, forgotten access cards and electronic failure.
- **User needs:** Secure locking, convenient authorised access and an independent backup method.
- **Design requirements:** Strong locking parts, simple operation, compact construction and protection against outdoor exposure.

Existing Gaps & Challenges

- Conventional locks may require a physical key and provide limited access flexibility.
- Electronic locks can become unusable if the battery or controller fails.
- Keys and RFID tags can be lost or forgotten.
- Outdoor locks must withstand moisture, vibration, repeated use and attempted tampering.
- The mechanical backup must remain independent of the electronic locking system.

Our Proposed Solution

Smart Bicycle Lock

A bicycle security system that combines electronic authentication with an independent mechanical backup, so users have an alternative way to unlock the bicycle if electronic access becomes unavailable.

- **Fingerprint and RFID:** Two convenient electronic access options for authorised users.
- **Mechanical backup:** A personalised left-right rotary sequence releases the locking bolt without battery power or controller operation.
- **Tamper alert:** A motion sensor detects suspicious movement and activates an audible alarm.

Designed for Everyday Parking

- Secure the bicycle to a suitable fixed object.
- Unlock using an authorised electronic method.
- Use the mechanical backup during electronic or battery failure.
- Use the additional attachment extension for more flexible parking options.

Technical Approach

Fingerprint Sensor / RFID Reader $\longrightarrow$ Microcontroller

$\Downarrow$

Actuator $\longrightarrow$ Locking Bolt

Motion Sensor $\longrightarrow$ Controller $\longrightarrow$ Buzzer

Rotary Combination $\longrightarrow$ Independent Mechanical Release

Electronic access controls the actuator, while the mechanical combination provides independent unlocking without electrical power.

Tech Stack

Frontend:	Not required for basic operation
Database:	Local credential storage, if supported
Backend:	Embedded control firmware
Hardware:	Fingerprint sensor, RFID reader, controller, actuator, motion sensor and buzzer

Innovation & Uniqueness

- **Dual electronic authentication:** Fingerprint and RFID provide convenient access options for authorised users.
- **Independent mechanical backup:** A personalised left-right rotary sequence releases the locking bolt without relying on the battery, motor or electronic controller.
- **Universal attachment extension:** An additional extension allows users to secure the bicycle to suitable poles, railings or other fixed structures, offering greater flexibility across different parking locations.
- **Integrated tamper detection:** A motion sensor and audible alarm provide an additional warning when suspicious movement is detected.
- **Combined design:** Electronic access, power-independent recovery, tamper awareness and flexible attachment are brought together in one bicycle lock.

Feasibility & Scalability

- **Technical feasibility:** RFID, fingerprint sensors, controllers, motion sensors and actuators are available for prototype integration.
- **Mechanical feasibility:** CAD, 3D printing and metal fabrication can produce the housing, bolt, rotary backup and attachment extension.
- **Independent backup:** The mechanical combination must release the bolt without electronics; verify this with the battery disconnected.
- **Materials and protection:** Evaluate steel for the bolt and aluminium alloy or other suitable housing materials. Test seals, connections and water resistance outdoors.
- **Risks and validation:** Test locking strength, repeated cycles, mechanical wear, false alarms, battery endurance, moisture ingress and attachment security before claiming reliability.
- **Scalability:** Start with campus trials, then standardise replaceable modules, mounting options and fabrication before considering larger production.

Cost for Product (Optional)

Component	Estimated Cost (Rs.)
ESP32 Development Board	373
RFID RC522 Reader with Card/Tag	139
AS608 Fingerprint Sensor	913
SG90 Servo Motor	155
SW-420 Vibration Sensor	38
18650 Rechargeable Battery (2200 mAh)	155
TP4056 Charging and Protection Module	22
18650 Battery Holder	17
Buzzer, LED and Resistors	50
Steel Locking Bolt, Spring and Guides	220
Mechanical Rotary Backup Mechanism	180
3D-Printed Housing and Metal Mount	250
Wires, Fasteners, Seals and Connectors	120
Fabrication and Testing Allowance	150
Total Estimated Cost	**2,782**

Cost Note

Costs are preliminary estimates and may vary with supplier, material quality and fabrication requirements. The servo is for prototype demonstration; a stronger locking actuator may be required for the final product.

Use Cases

- **Use case 1 – Campus parking:** A student secures a bicycle to a suitable fixed object using the lock and unlocks it with authorised fingerprint or RFID access.
- **Use case 2 – Electronic failure:** If the battery is depleted or electronic access fails, the owner uses the correct left-right rotary combination to release the bolt mechanically.
- **Use case 3 – Tampering:** Unexpected movement of a secured bicycle triggers the motion sensor and local audible alarm.

Impacts and Benefits

- **Convenient access:** Fingerprint and RFID authentication provide electronic alternatives to carrying a conventional key.
- **Access during failure:** The independent mechanical backup offers another unlocking method when the battery or electronics fail.
- **Tamper awareness:** A motion sensor and audible alarm alert nearby people when suspicious movement is detected.
- **Flexible parking:** The attachment extension allows the bicycle to be secured to suitable fixed structures at different parking locations.
- **Evidence-based improvement:** Prototype testing can measure access reliability, backup operation, alarm response, battery life, weight and resistance to outdoor conditions.

Sustainable Development Goals (SDGs)

SDG	Target	Goal Name	Project Connection
SDG 9	Target 9.5	Industry, Innovation and Infrastructure	Develops an integrated bicycle lock by applying engineering and technological innovation to electronic access, mechanical backup and tamper detection.
SDG 11	Target 11.2	Sustainable Cities and Communities	Addresses a bicycle-parking security concern and may support everyday cycling; increased cycling or improved transport access would need separate evidence.

These are intended areas of alignment, not claims of measured SDG impact. Official references: UN SDG 9 and UN SDG 11.

Individual Member Contribution

Member	Contribution or responsibility
Shree Ishwarya R	Team lead, problem definition, design coordination, system planning and presentation.
Sridhar V R	Mechanical design, CAD modelling, locking structure, housing and bicycle mounting.
Saai Adith K	Prototype assembly, testing, evaluation and design refinement.
Roopali S	Existing-solution research, user feedback, documentation and supporting research.

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